Lecture Notes on

The Edges of Time

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These notes summarize the thirteenth and final Astrophysics Club meeting. Everyone knows the Big Bang; we skip past it. The interesting question is what we *don't* know – what came before the Big Bang, and what comes after our universe ends. Both mysteries live in the same regime of physics we do not yet possess.

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1 Two Questions at the Edge of Time

The Big Bang gives us a familiar reference point: *now* sits roughly 13.8 Gyr after that event. Everything on either side is the subject of this meeting.

- **Before the beginning.** What happened, if anything, before the universe existed?
- **After the end.** How does the story finish?

Both questions are governed by the same equations at opposite ends of time. Both depend on physics we do not yet have.

2 Part I: Before the Beginning

2.1 What We Know

The universe is approximately 13.8 billion years old, as inferred from the cosmic microwave background. The current best model of the first moments is **inflation**: an exponential expansion of space in roughly the first 10^{-32} seconds after the Big Bang.

Inflation elegantly explains several long-standing puzzles:

- The smoothness of the CMB across the entire sky.
- The observed flatness of space.
- The absence of magnetic monopoles.
- The origin of the density fluctuations that seeded galaxies.

Inflation is well-supported by data, though the mechanism driving it is still not directly observed.

2.2 The Master Equation of Cosmology

Every scenario we discuss is a different solution to the same equation under different conditions – the Friedmann equation:

$$H^2 = \frac{8\pi G}{3}\,\rho \;-\; \frac{k}{a^2}.$$

Here H is the expansion rate, ρ is the energy density, k is the spatial curvature, and a is the scale factor. This is the master equation of cosmology. Every idea in this meeting is a different way of solving it near $t = 0$ or as $t \to \infty$.

2.3 What We Don't Know

- What caused inflation? The driving field, called the *inflaton*, is hypothesized but not confirmed.
- Was there anything before inflation at all?
- This is exactly where known physics breaks down.

Inflation is usually modeled by a scalar field ϕ that rolls down a potential $V(\phi)$. Its equation of motion is

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0.$$

The rolling of ϕ down $V(\phi)$ is what drives the exponential expansion. The catch: we do not know the shape of $V(\phi)$.

3 Leading Ideas for Before the Beginning

There are several serious proposals for what preceded (or replaces) the Big Bang. Each is a different solution to the master equation under different assumptions.

3.1 Time Began at the Big Bang (Hawking)

There is no "before." Asking what happened before the Big Bang is like asking what is north of the North Pole – the question is well-formed in language but not in physics. Time itself started at $t = 0$.

3.2 Eternal Inflation (Linde, Vilenkin, Guth)

Inflation never fully stops. Our universe is a single "bubble" that stopped inflating, inside a much larger inflating background that is still going. Other bubbles are still forming. Different bubbles can even have different effective laws of physics. This is one of the physics-based origins of the modern idea of a *multiverse*.

3.3 Cyclic Universe and Big Bounce (Steinhardt, Turok)

The Big Bang was not the beginning but a rebound. A previous universe contracted, reached a minimum size, and bounced back into expansion – our expansion. In this picture there is no beginning and no end, only an endless cycle.

3.4 Conformal Cyclic Cosmology (Penrose)

Each universe's heat death seeds the next. Penrose observed that at maximum cold (empty, in thermal equilibrium) and maximum hot (dense radiation), the conformal geometry seen by light looks the same. Aeons follow one after another, with each ending becoming another's beginning.

3.5 No-Boundary Proposal (Hartle and Hawking)

Spacetime is finite but has no edge – in the same way the surface of a sphere is finite but has no boundary. Time near the "beginning" behaves as an imaginary dimension, smoothly rounding off rather than terminating at a point. Asking for a "first moment" is like asking for a "first point" on the surface of the Earth.

4 Why Explore This?

Every idea in this section lives in the realm of **quantum gravity** – the unification of general relativity and quantum mechanics, a theory we do not yet have.

The underlying question is old and simple:

- Why is there something rather than nothing?

Solving the beginning of the universe means solving string theory, the multiverse, and the foundations of physics itself.

5 Part II: After the End

5.1 What We Know

Three facts anchor everything about the fate of the universe:

- The universe is expanding.
- The expansion is *accelerating*.
- The acceleration is driven by dark energy.

Which of the fate scenarios below actually plays out depends entirely on the nature of dark energy.

5.2 The Equation of State

Dark energy is characterized by its equation of state,

$$w = \frac{p}{\rho},$$

the ratio of its pressure to its energy density. For a vanilla cosmological constant, $w = -1$ exactly. Every fate scenario that follows corresponds to a different value (or a different history) of w.

This one number is the most consequential quantity in cosmology that we cannot yet measure precisely.

6 The Big Freeze

The current best-guess ending under standard ΛCDM is the *Big Freeze*. Galaxies isolate as space expands. Stars die. Black holes evaporate. Entropy wins.

Rough timeline:

10^{12} yr	Stars burn out
10^{36} yr	Protons decay (if they do)
10^{67} yr	Stellar-mass black holes evaporate
10^{100} yr	Last supermassive black holes evaporate

Black holes evaporate via Hawking radiation. The *Hawking temperature* of a black hole of mass M is

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}.$$

The temperature scales as $T_H \propto 1/M$, so more massive black holes are colder and evaporate more slowly. The last supermassive black holes take on the order of 10^{100} years to die out. After that, nothing capable of doing work remains anywhere.

7 Other Endings

7.1 Big Rip ($w < -1$)

Dark energy strengthens with time. Expansion accelerates so violently that it tears apart first galaxies, then solar systems, then atoms – in finite time. The end is not gradual; it is catastrophic and dated.

7.2 Big Crunch ($w > -1$)

Dark energy weakens or reverses. Gravity eventually wins, and the universe collapses back on itself, running the expansion history in reverse and ending in a state resembling the Big Bang.

7.3 Vacuum Decay (any w)

The current vacuum may not be the lowest-energy state. If it isn't, a bubble of true vacuum can nucleate somewhere in space and expand outward at the speed of light, rewriting the laws of physics inside. There is no warning; it could happen at any moment.

8 What We Don't Know

Which fate scenario is real is determined by dark energy.

- ΛCDM predicts $w = -1$, constant. For decades, this was the accepted answer.
- The 2024 DESI results hint that w is evolving with time.
- If those results hold, several fate scenarios that ΛCDM had ruled out come back into play.

A common way to parameterize evolving dark energy is

$$w(a) = w_0 + w_a(1 - a),$$

where a is the scale factor. w_0 describes dark energy's strength today, and w_a describes how it changes over cosmic time. DESI's 2024 measurements find, approximately,

$$w_0 \approx -0.7, \qquad w_a \approx 1.0.$$

If these numbers hold up, we are, right now, in the middle of potentially revising the ending.

9 Why It Matters

Every mystery we have covered this year feeds into this one question.

- Dark matter
- Dark energy
- Black holes
- Exoplanets
- Stellar evolution
- Special relativity
- Cosmology
- Fundamental constants

Each of these enters somewhere in the story of how the universe began and how it will end.

10 The Connection

The two mysteries may not really be two mysteries.

The Beginning	infinitely hot, infinitely dense, before time.
The End	infinitely cold, infinitely empty, after time.

The same physics – physics we do not yet have – governs both. The same ignorance that keeps us from knowing what came before keeps us from knowing what comes after.

11 Closing

This is the last regular meeting of the club for the year. The intent of this final session was not to close the year with an answer, but with a question:

- What lies before the beginning?
- What lies after the end?

Both are open. Both are worth spending a lifetime on. Thanks for coming to all our meetings this year.